JLPT AI App, Progress Report No. 03, HP

Padauk Lab

Report	Period	Date	Prepared by
No. 03	Oct 13, 2026	Oct 13, 2026	HP (Backend and Infrastructure)

You can share this file with your AI coding tool as project context, together with docs/api-contract.md and docs/database-design.md.

1. Summary

The API contract is merged into main. The database is designed and built: **27 tables in 5 groups**, written as Python models, with the first migration. All 27 tables were created and checked in PostgreSQL in Docker. The design (pull request #4) and the code (pull request #5) are waiting for review.

2. Completed this period

No.	Task	Owner	Status
1	API contract (pull request #2) reviewed and merged into main	HP, team	Done
2	Database design v0.1 in docs/database-design.md (27 tables, 5 groups)	HP	In review (pull request #4)
3	New libraries added and pinned: SQLAlchemy, Alembic, psycopg	HP	In review (pull request #5)
4	Database connection (backend/app/db.py), reads DATABASE_URL from .env	HP	In review (pull request #5)
5	Python models for all 27 tables (backend/app/models/)	HP	In review (pull request #5)
6	Alembic set up, and the first migration that creates every table	HP	In review (pull request #5)
7	Tested in Docker: 27 tables created, code and database match	HP	Done

3. The database at a glance

Group	Tables	What it stores
1. Users and login	5	Accounts, login tokens, password reset links, devices, notification settings
2. Study setup and plan	3	Setup answers, study plan, official exam dates
3. Learning content	6	Sessions, units, words, grammar, example sentences, proverbs
4. Quizzes and mock tests	4	Quizzes, mock tests, reading passages, questions
5. Progress and AI	9	Units done, saved words, quiz and mock test attempts and answers, saved Why? explanations, AI usage

- **Reset all progress** adds 1 to the user's round. Nothing is deleted, so old rounds stay as history.
- **Calculated, not stored**: days left, progress, best score and pass prediction are worked out when needed, so they are never out of date.
- **Google login** uses Google's permanent user ID (google_sub) in the users table. No extra table is needed.

- **Furigana** is stored twice: as plain text for search (furigana) and in the contract's ruby format for display.

4. Safety rules inside the database

The database itself refuses bad data, even if the code has a bug. Examples:

Rule	What it blocks
An account needs a password or a Google ID	Accounts nobody can log in to
Emails are stored lowercase	Two accounts for Htet@x.com and htet@x.com
No score when the last level is not taken	Impossible setup answers
Only one mock test in progress per user and test	Two open tests at once (retakes after submitting are allowed)
Every question has exactly 4 options, answer a to d	Broken questions
One answer per question per quiz attempt	Answering twice
Deleting a user deletes their data	Personal data left behind
Content can't be deleted while something uses it	Losing words or questions by accident

Passwords and tokens are stored only as hashes. Correct answers are never sent to the app before the user answers.

5. How to get the database on your computer

After pull request #5 is merged:

1. `git checkout main` and `git pull`
2. `docker compose run --rm --build backend alembic upgrade head`
3. Check with `\dt` in psql (see the pull request): 28 rows, which is 27 tables plus alembic_version

The database starts empty. Content from ANK is loaded in later.

6. What everyone should do now

1. Review pull requests #3, #4 and #5. One approval each is enough
2. In pull request #4, check your part. ANK: Groups 3 and 4 (content and questions). SWYH: ai_explanations and ai_usage. LMHP: anything that does not fit the screens
3. If something in the design must change, comment in pull request #4 now. Later changes are possible, but each one needs a new migration

7. Next tasks (Week 1, Oct 12 to 18)

No.	Task	Owner	Depends on
1	Review pull requests #3, #4 and #5	LMHP, SWYH, ANK	Nothing
2	Placeholder endpoints that return the contract's example JSON, so the app can connect early	HP	Nothing
3	Real sign up and login endpoints (Section 1), with password hashing	HP	Task 1 (#5 merged)
4	Design system and onboarding and dashboard screens, using the contract's examples as fake data	LMHP	Nothing
5	One complete sample unit in the contract format	ANK	Nothing

No.	Task	Owner	Depends on
6	First proverb list (effort, consistency, never giving up, not wasting time)	ANK	Nothing
7	Compare LLMs, then agree with HP on the AI functions for More examples and Why?	SWYH	Nothing

8. Open decisions

1. Pass prediction formula (HP and ANK to propose)
2. Exact question split in the mock test template (ANK to confirm against the N1 format)
3. How last level and score change the daily pace (HP to propose)
4. Which LLM to use (SWYH to propose)
5. Daily limit for More examples, currently 50 (team to decide after testing)
6. Quiz question bank size, so retakes get a different mix (ANK, suggestion: 40 per quiz)
7. When content is corrected, should saved Why? explanations for that question be regenerated? (HP and SWYH, suggestion: yes)
8. If someone signs up with email, then later uses Google with the same email, join the accounts or refuse? (HP to propose. Join only when Google says the email is verified)